Lowcountry Shrimp Dataset Deep Dive

July 30, 2025

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Summary

About the Project

Funded by the NERRS Science Collaborative, link to page

Collaborators

Name people here!

Part I

Definitions and Structure

1 Shrimp Year Abundance Index Explanation

The concept of **shrimp year** is somewhat complicated - a detailed explanation is below. If you prefer, jump straight to Figure 1.1.

1.1 Shrimp Year

When examining long-term datasets, it is convenient, if not necessary, to perform some aggregation of data to means or totals over some representative length of time (e.g. monthly mean temperature, annual precipitation total, monthly or annual totals of money spent on [anything related to budgets]).

In this project, we want to be able to link different life stages of shrimp to each other, and ask questions like "if postlarvae did well this year, did juveniles also do well? does that hold all the way through adults, and fisheries landings?"

Animal life cycles, incoveniently for us, do not necessarily align with calendar years. It may be (and for brown shrimp generally is the case) that young are spawned early in a calendar year and are adults late in that calendar year. However, white shrimp (and many other species) may be spawned in the middle or late part of a calendar year, and not be adults until the next calendar year. So, in order to relate those adults to the earlier life stages, we cannot use calendar year to represent the cohort of shrimp.

We use **shrimp year** to represent a single cohort of shrimp, and assign the calendar year in which that group of shrimp will live most of their lives. For brown shrimp, the shrimp year is the calendar year in which the individuals will show up in the fishery. For white shrimp, it is the year in which they go through most life stages; though adults will bleed into (and spawn in) the following calendar year.

Each survey in this project, based on its location of sampling, is expected to primarily catch a single life stage of each shrimp. Based on the month of sampling, the targeted life stage, and the shrimp species, we assigned shrimp year to each sampling event based on the following:

insert tables here of shrimp year definitions by life stage

1.1.1 Environmental data

Winter temperatures and cold spells can impact shrimp growth and/or survival, and understanding these dynamics is important to shrimp fishery managers. Salinity during early life stages may also impact shrimp, and these effects extend to later stages of the life cycle.

Again, we need to aggregate data. **Winter** here is defined as December through February. For brown shrimp, the shrimp year impacted by winter temperatures is the calendar year containing that January and February. For white shrimp, we have assigned each winter metric separately to two life stages, as they occur in different years: postlarval winters and adult winters. Subadults and adults will be impacted by winter temperatures occurring a year after the winter temperatures that affect the postlarval and juvenile stages. See Figure 1.1. **Nursery period** is defined separately for each species, based on Batchelder et al. 2024*: April through July for brown shrimp, and June through October for white shrimp.

*Batchelder, L., Stone, J., Kimball, M., Pfirrmann, B., & Dunn, R. (2024). Phenology of penaeid shrimp nursery habitat use: Trends and environmental drivers over four decades. Marine Ecology Progress Series, 751, 79–95. https://doi.org/10.3354/meps14741

1.2 Graphic of abundances, temperatures, and assigned shrimp years

this will be what Kim showed the group at some point in the past, with 2 or 3 calendar years worth of data

need to actually process the data though

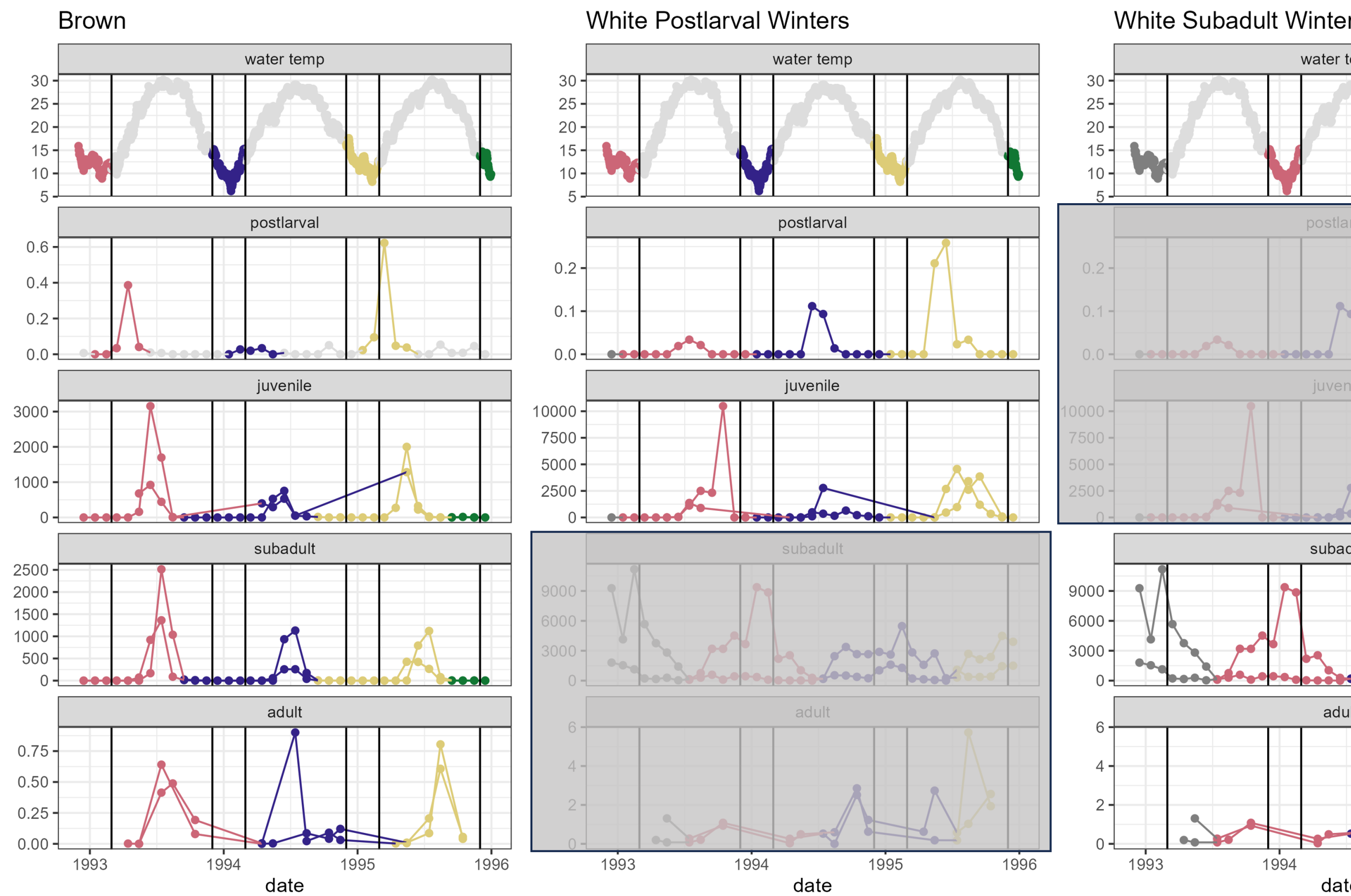


Figure 1.1: Temperature and abundance data, colored by shrimp year, with irrelevant life stages grayed out.

group may want to insert conceptual diagram too when that is completed

1.3 Abundance Index

This project brought together a number of surveys by different organizations, with different methods, and different ways of quantifying shrimp abundance. In order to make these numbers comparable, while accounting for possible variation in the number of sampling stations and sample dates over time, we have adjusted abundance metrics to an index using the following steps. More details for individual datasets will be provided in that life stage's chapter in the "Summarizing to Abundance Index" portion of this book.

For each survey:

1. Assign shrimp year to each sample, based on month and species.
2. If a sampling event (site/station on a date) had replicates, average them.
3. Average data across all sites/stations on a sample day.
4. Average data across all sampling dates within each month = average abundance for the month.
5. Add the monthly averages, to get a total abundance for each shrimp year. This absorbs variability in timing for a year.
6. From total abundance per shrimp year, derive an index:
    a. Take the square root of total abundance (deals some with extreme high numbers).
    b. Mean-center the result from (1) across all shrimp years for the dataset.
    c. Divide by the standard deviation across all shrimp years for the datset.

talk about multiple datasets and how they were combined

2 Data Processing and Directory Structure

This chapter primarily exists to guide other members of the working group in modifying, updating, or extracting information from this book.

2.1 Directory organization

- `_quarto.yml` is the main 'table of contents' of all .qmd files in order. Each chapter also has an 'About' section at the bottom identifying the exact file used to generate it, as well as details about the input and output datasets for the chapter.

- `_book/` is the subfolder with output files. Open any .html in here and you'll have access to the full structure. The pdf output is also in this directory.

2.2 Data, data, data

- `data/raw` = time series, unmodified

- `data/intermediate` = time series with some simplifications for reporting and summarizing; generally outputs from the "Data Exploration" chapters of this book

- `data/processed` - as of this writing, I haven't processed data yet. This folder will contain data that are aggregated to different levels:

- month or season

- shrimp year (based on month or season); these files will also contain a column for the abundance index

The older version of the processed data folder has the following subfolders:
- *01 Survey level* - averaged to month and then shrimp year, per survey. ALL files have the same columns: survey, species, life_stage, shrimp_year, abundance_measure, abundance, sqrt_abundance
- *02 Life Stage level* - averaged survey results to shrimp year x life stage, if relevant Again

all files have same columns. Survey, species, life_stage, shrimp_year, abundance, abundance_measure (if average of two surveys, will be told here), abundance, sqrt_abundance
- *03 Combined abundance* - combined all abundance files and calculated abundance index
- *04 Environmental Survey level*
- *05 Combined environmental*

2.3 About

Input `.qmd` file for this section was: **directory_structure.qmd** and can be found in the main directory.

Part II

Shrimp Data Exploration

Here, we go through each file to see the distribution of data and to make some graphics illustrating what is present in each dataset.

3 Postlarvae

3.1 Dataset

The only dataset representing postlarvae is from North Inlet-Winyah Bay.

3.2 Graphics - Size Distributions

In these plots, there is a panel for each shrimp species. Size is represented on the y-axis, and a cloud of points called a "beeswarm" has representation for every data point. Each point is colored by the numeric month in which the individual was caught, allowing us to see differences in size at different parts of the year and life-cycle.

There are three different measures of size in this dataset: Rostrum length, Telson length, and Uropod length. A graph is below for each.

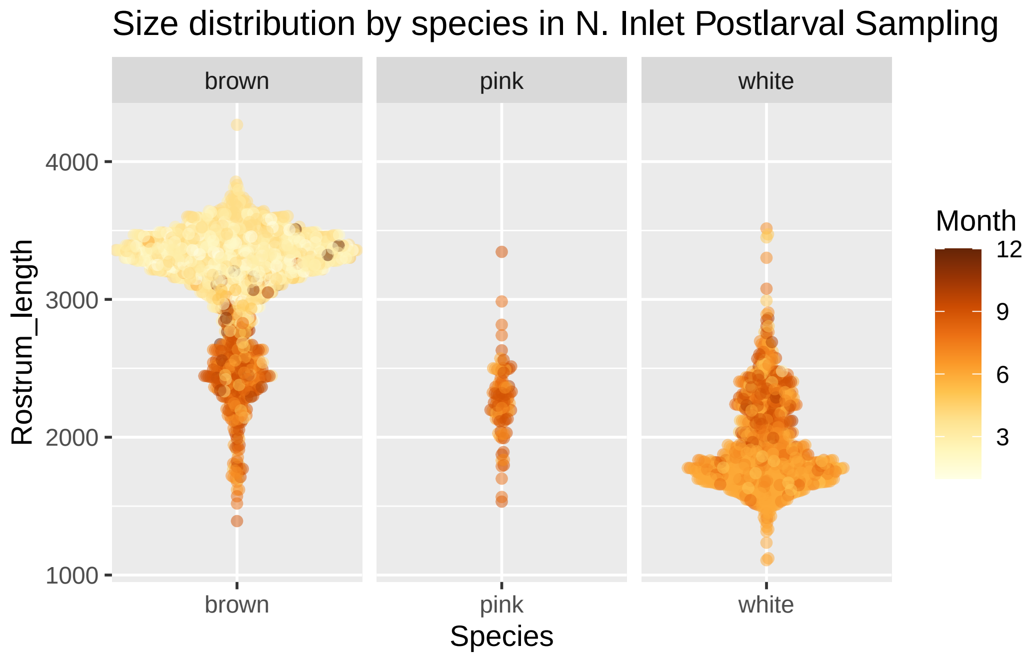


Figure 3.1: Beeswarm plot of rostrum length in postlarval shrimp. Points represent measurements of individual shrimp, and are colored by the month in which they were captured. The y-axis represents length. More points spread out along the x-axis for a given length means that there were more individuals of that length captured than regions where points remain closer to the center.

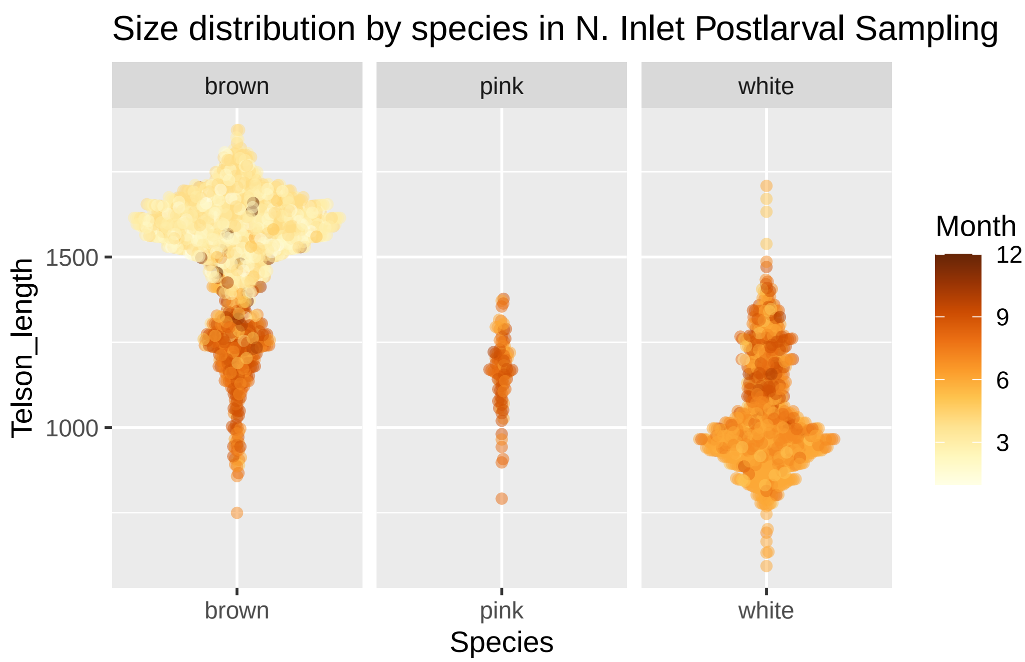


Figure 3.2: Beeswarm plot of telson length in postlarval shrimp. Points represent measurements of individual shrimp, and are colored by the month in which they were captured. The y-axis represents length. More points spread out along the x-axis for a given length means that there were more individuals of that length captured than regions where points remain closer to the center.

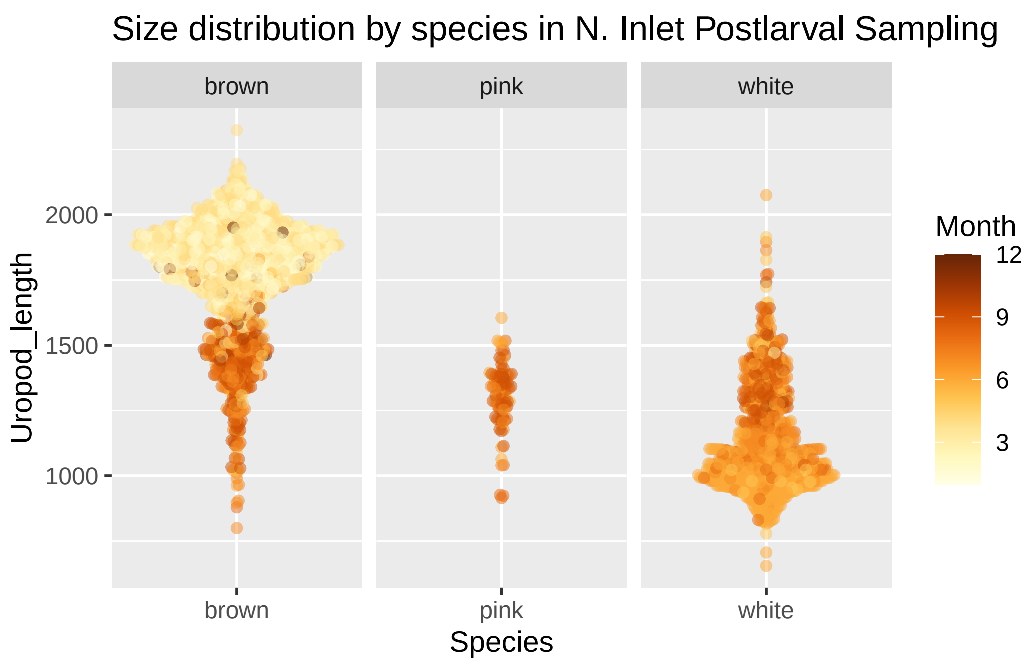


Figure 3.3: Beeswarm plot of uropod length in postlarval shrimp. Points represent measurements of individual shrimp, and are colored by the month in which they were captured. The y-axis represents length. More points spread out along the x-axis for a given length means that there were more individuals of that length captured than regions where points remain closer to the center.

With brown shrimp, looks like we've got a couple of size classes in here - bigger shrimp earlier in the year, and smaller ones that are mostly later months.

The plot below shows the entire time series, with date (as year-month) along the x-axis and uropod length on the y-axis. Points are again colored by month.

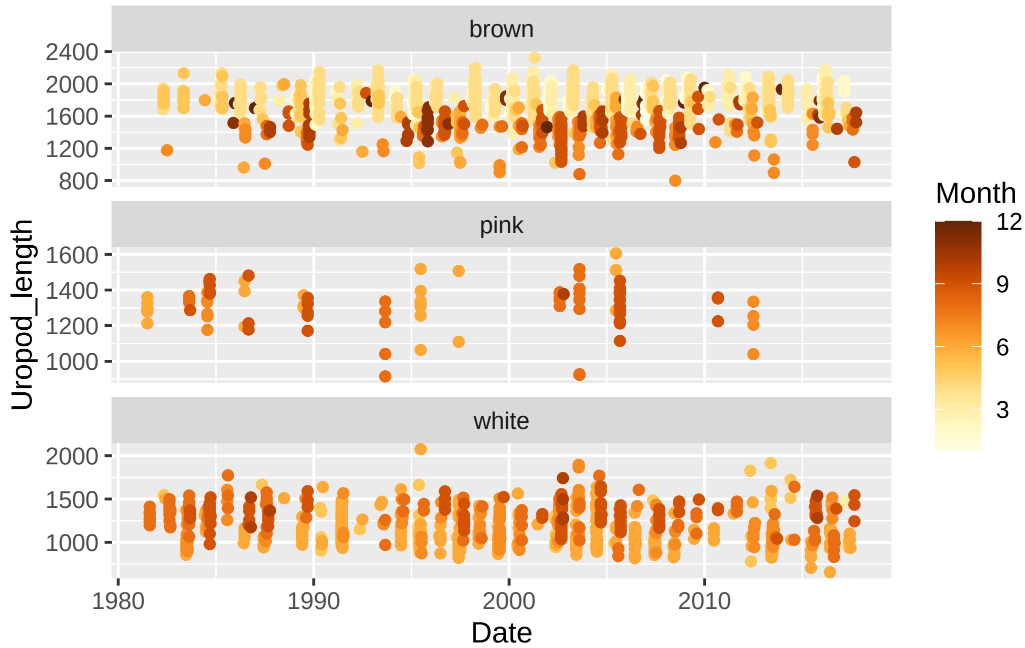


Figure 3.4: Time series plot of uropod length in postlarval shrimp. Points represent measurements of individual shrimp, and are colored by the month in which they were captured.

3.3 Graphics - Abundance and Size Boxplots

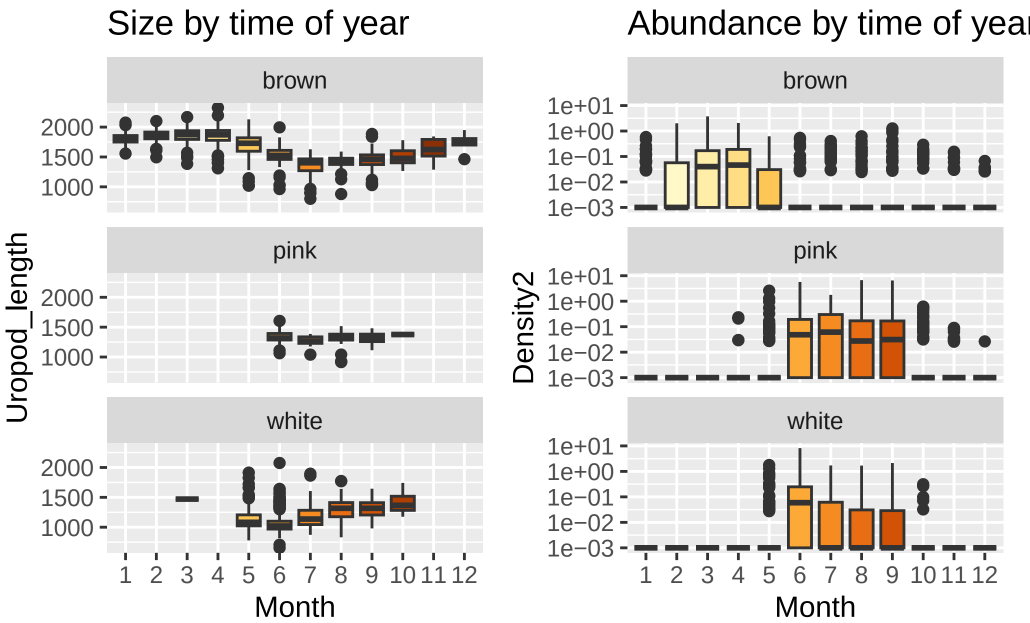


Figure 3.5: Boxplots representing shrimp size and abundance Note the log10-scaled y-axes.

So even though we're seeing that pattern in brown shrimp of being so much bigger early in the year and smaller later, they're just really not all that common later in the year.

3.4 Tabular summary of abundance

Table 3.1: Abundance data frame summary

Table 3.1: Data summary

Name	postlarv_abund
Number of rows	1830
Number of columns	21
Column type frequency:	
character	6
Date	1

numeric	14
Group variables	None

Variable type: character

Table 3.2: Abundance data frame summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
replicate	0	1	1	10	0	13	0
date	0	1	0	10	6	915	0
season	0	1	4	6	0	4	0
ol_sal	0	1	0	7	2	181	0
ol_temp	0	1	0	7	2	241	0
notes	0	1	0	29	1746	22	0

Variable type: Date

Table 3.3: Abundance data frame summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
date2	6	1	1981-01-20	2017-12-30	1999-06-17	914

Variable type: numeric

Table 3.4: Abundance data frame summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
sample	0	1	458.00	264.21	1.0	229.25	458.00	686.75	915.00
year	0	1	1998.67	12.07	1900.0	1990.00	1999.00	2008.00	2017.00
month	0	1	6.48	3.45	1.0	3.00	6.00	9.00	12.00
day	0	1	15.58	8.83	0.0	8.00	16.00	23.00	31.00
dayofyear	0	1	181.63	105.57	0.0	90.25	181.00	271.75	365.00
week	0	1	26.96	15.08	1.0	14.00	27.00	40.00	53.00
bb_surface_salinity	0	1	31.59	5.08	5.9	30.83	33.20	34.50	38.60
bb_surface_temp	0	1	19.28	7.15	3.3	13.20	19.50	26.37	33.00
total_153_zoop_ind_m.3	0	1	9946.45	10692.28	0.0	3440.03	6531.50	12306.58	119401.90
vol.filt.0.13	7	1	32.36	4.76	0.0	29.36	32.47	35.74	45.68
total_ppl_density	9	1	0.20	0.63	0.0	0.00	0.00	0.13	8.40

Table 3.4: Abundance data frame summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
brown_density	9	1	0.05	0.19	0.0	0.00	0.00	0.03	3.77
white_density	9	1	0.05	0.29	0.0	0.00	0.00	0.00	8.19
pink_density	9	1	0.09	0.42	0.0	0.00	0.00	0.00	6.71

3.5 Tabular summaries of size

3.5.1 Brown Shrimp

Table 3.5: Brown shrimp size summary

Table 3.5: Data summary

Name	postlarv_size_brown
Number of rows	1952
Number of columns	16
Column type frequency:	
character	5
numeric	10
POSIXct	1
Group variables	None

Variable type: character

Table 3.6: Brown shrimp size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Shrimp_ID	0	1.00	6	8	0	1949	0
TowID	0	1.00	3	4	0	414	0
Replicate	0	1.00	1	1	0	2	0
Species	0	1.00	5	5	0	1	0
Notes	1830	0.06	1	34	0	33	0

Variable type: numeric

Table 3.7: Brown shrimp size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Cruise	0	1.00	510.58	192.75	32.00	388.00	521.00	606.00	909.00
Shrimp_ID_number	0	1.00	10.36	13.36	1.00	2.00	5.00	14.00	73.00
Surface_Salinity	2	1.00	26.58	7.84	5.90	23.10	29.60	32.38	38.60
Surface_Temperature	2	1.00	17.30	5.44	7.00	13.60	16.50	19.50	30.10
Rostrum_teeth	40	0.98	2.68	0.84	0.00	2.00	3.00	3.00	6.00
Rostrum_length	61	0.97	3141.90	415.89	1391.45	3018.81	3278.33	3415.37	4266.98
Telson_length	70	0.96	1516.31	179.36	749.62	1441.84	1567.86	1635.95	1872.39
Uropod_length	38	0.98	1768.37	214.82	799.73	1675.06	1822.97	1914.16	2324.15
Year	0	1.00	2001.33	7.81	1982.00	1996.00	2001.50	2005.00	2017.00
Month	0	1.00	4.26	2.31	1.00	3.00	4.00	4.00	12.00

Variable type: POSIXct

Table 3.8: Brown shrimp size summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1982-04-20	2017-10-03	2002-01-03	302

3.5.2 White Shrimp

Table 3.9: White shrimp size summary

Table 3.9: Data summary

Name	postlarv_size_white
Number of rows	1096
Number of columns	16
Column type frequency:	
character	5
numeric	10
POSIXct	1
Group variables	None

Variable type: character

Table 3.10: White shrimp size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Shrimp_ID	0	1.00	7	8	0	1087	0
TowID	0	1.00	3	4	0	249	0
Replicate	0	1.00	1	1	0	2	0
Species	0	1.00	5	5	0	1	0
Notes	1029	0.06	5	35	0	29	0

Variable type: numeric

Table 3.11: White shrimp size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Cruise	0	1.00	487.77	228.39	15.00	356.00	482.00	632.50	907.00
Shrimp_ID_number	0	1.00	7.28	8.19	1.00	2.00	4.00	10.00	47.00
Surface_Salinity	0	1.00	31.57	4.82	6.90	31.40	33.20	34.40	38.60
Surface_Temperature	0	1.00	26.03	2.43	13.90	25.20	26.70	27.50	30.10
Rostrum_teeth	16	0.99	1.65	0.79	0.00	1.00	2.00	2.00	6.00
Rostrum_length	27	0.98	1940.35	321.57	1107.64	1710.19	1831.93	2179.20	3514.92
Telson_length	28	0.97	1033.78	155.01	593.65	928.25	990.63	1144.55	1708.58
Uropod_length	18	0.98	1142.02	197.06	654.82	995.32	1081.15	1281.64	2075.20
Year	0	1.00	2000.24	9.24	1981.00	1995.00	2000.00	2006.00	2017.00
Month	0	1.00	6.65	1.16	3.00	6.00	6.00	7.00	10.00

Variable type: POSIXct

Table 3.12: White shrimp size summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1981-08-13	2017-09-01	2000-06-29	184

3.6 About

Input `.qmd` file for this section was: **raw_postlarv.qmd** and can be found in the main directory.

Input data file(s) - as-used; can be found in `data/raw/postlarv`:

- `Penaeus_PostLarvae_NInlet_1981_2017_wide_kac.csv`
- `Penaeus_Postlarval_Lengths_NInlet_1981 - 2017.xlsx`

Modifications to data:

- minor re-naming: inserted underscores rather than spaces; changed `Shrimp Species` to `Species`.

Generated data file(s): Data frames were modified as described above, and written to `data/intermediate/postlarval_dfs.RData` (the single `.RData` file contains multiple data frames):

- `postlarv_abund`
- `postlarv_size`

Other notes about data:

none

4 Juveniles

4.1 Datasets

There are two datasets that represent juveniles: SCDNR's Creek Trawls, and NIW NERR's Oyster Landing seines. We will explore each.

4.2 Graphics - Size Distributions

Beeswarm plots can be slow to render so I have subsetted both data frames here to 15,000 rows.

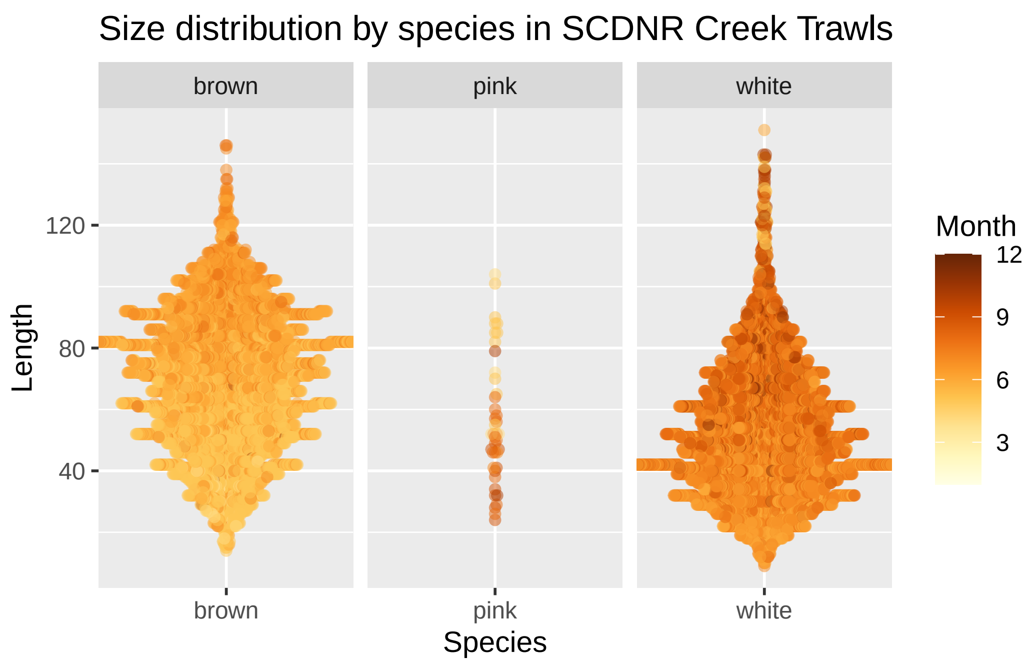


Figure 4.1: Beeswarm plot of juvenile shrimp length for the SCDNR data. Points represent measurements of individual shrimp, and are colored by the month in which they were captured. The y-axis represents length. More points spread out along the x-axis for a given length means that there were more individuals of that length captured than regions where points remain closer to the center.

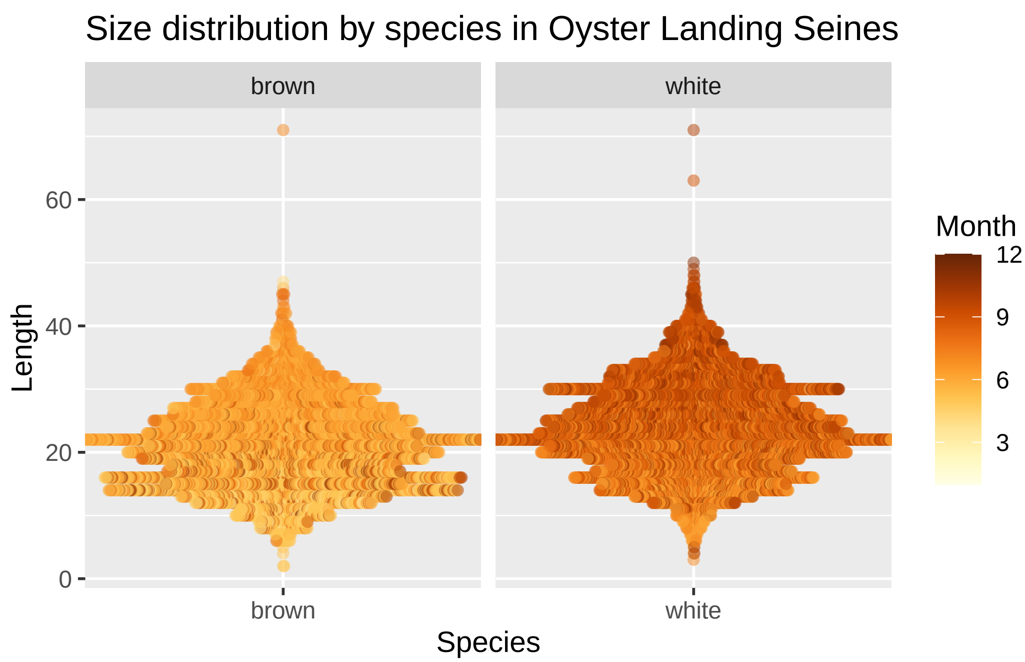


Figure 4.2: Beeswarm plot of juvenile shrimp length for the Oyster Landing seine data. Points represent measurements of individual shrimp, and are colored by the month in which they were captured. The y-axis represents length. More points spread out along the x-axis for a given length means that there were more individuals of that length captured than regions where points remain closer to the center.

Looks like OL seines are generally catching smaller individuals of both species than the SCDNR creek trawls.

4.3 Graphics - Abundance/CPUE Boxplots

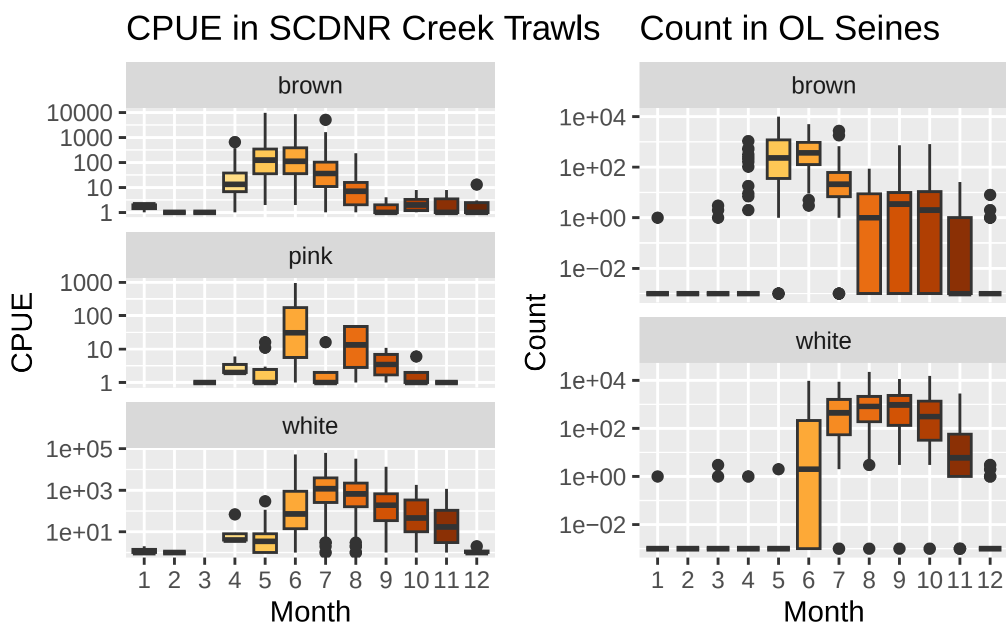


Figure 4.3: Boxplots representing shrimp abundance (as either CPUE or count, depending on the survey) by month. Note the log10-scaled y-axes.

Both surveys (unsurprisingly) show the same temporal pattern of abundance - aztecus in May/June, with setiferus later in the year.

4.4 Tabular summaries of abundance

4.4.0.1 SCDNR Creek Trawls

Table 4.1: Abundance (SC) data frame summary

Table 4.1: Data summary

Name	juv_sc_cpue
Number of rows	5619
Number of columns	13

Column type frequency:

character	5
numeric	8
Group variables	None

Variable type: character

Table 4.2: Abundance (SC) data frame summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
StationCode	0	1	4	4	0	7	0
EstuaryCode	0	1	2	2	0	1	0
DTStart	0	1	8	10	0	621	0
SpCode	0	1	4	4	0	3	0
Species	0	1	4	5	0	3	0

Variable type: numeric

Table 4.3: Abundance (SC) data frame summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
X	0	1	2810.00	1622.21	1.00	1405.50	2810.00	4214.50	5619.00
Coll	0	1	20030686.43	173365.28	19801003.00	19911032.00	20031007.00	20161081.00	20231084.00
Year	0	1	2002.97	13.34	1980.00	1991.00	2003.00	2016.00	2023.00
Month	0	1	6.62	1.91	1.00	5.00	7.00	8.00	12.00
Day	0	1	15.29	8.38	1.00	9.00	15.00	22.00	31.00
CPUE	3	1	546.00	2690.49	0.00	0.00	0.00	82.00	62976.00
Lat	0	1	32.84	0.07	32.75	32.80	32.86	32.86	32.95
Long	0	1	-79.89	0.08	-79.99	-79.98	-79.88	-79.85	-79.76

4.4.0.2 Oyster Landing Seines

Table 4.4: Abundance (Oyster Landing) data frame summary

Table 4.4: Data summary

Name	juv_ol_count
Number of rows	1796
Number of columns	12

Column type frequency:	
character	4
Date	1
numeric	7
Group variables	None

Variable type: character

Table 4.5: Abundance (Oyster Landing) data frame summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Protocol	0	1	7	8	0	3	0
WQ.Source	2	1	8	19	0	4	0
WQ.Time	2	1	0	5	36	74	0
Species	0	1	5	5	0	2	0

Variable type: Date

Table 4.6: Abundance (Oyster Landing) data frame summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1984-01-04	2023-12-21	2002-02-18	898

Variable type: numeric

Table 4.7: Abundance (Oyster Landing) data frame summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Sample	0	1.00	449.76	259.63	1.0	225.0	449.5	675.00	899.0
Year	0	1.00	2002.63	11.71	1984.0	1993.0	2002.0	2012.00	2023.0
Month	0	1.00	6.60	3.26	1.0	4.0	7.0	9.00	12.0
Temp	94	0.95	20.47	6.91	3.8	14.5	21.6	26.90	31.4
Sal	106	0.94	31.82	4.78	2.3	30.0	33.3	35.00	38.7
Weight	489	0.73	625.27	2355.88	0.0	0.0	0.0	77.05	39196.1
Count	240	0.87	380.32	1326.23	0.0	0.0	0.0	62.00	22560.0

4.5 Tabular summaries of size

4.5.1 Brown Shrimp

4.5.1.1 SCDNR Creek Trawls

Table 4.8: Brown shrimp (SC) size summary

Table 4.8: Data summary

Name	sz_sc_brn
Number of rows	34818
Number of columns	14
Column type frequency:	
character	4
Date	1
numeric	9
Group variables	None

Variable type: character

Table 4.9: Brown shrimp (SC) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
DTStart	0	1	8	10	0	489	0
StationCode	0	1	4	4	0	7	0
SpCode	0	1	4	4	0	1	0
Species	0	1	5	5	0	1	0

Variable type: Date

Table 4.10: Brown shrimp (SC) size summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1980-05-08	2023-12-11	1996-06-12	489

Variable type: numeric

Table 4.11: Brown shrimp (SC) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
X	0	1.00	35073.55	22862.74	1.00	15867.25	32821.50	52650.75	81763.00
Coll	0	1.00	19981663.40	119955.91	19801003.00	19891005.00	19961014.00	20061026.00	20231080.00
Length	0	1.00	69.54	22.67	12.00	52.00	71.00	86.00	151.00
Lat	0	1.00	32.83	0.07	32.75	32.80	32.81	32.86	32.95
Long	0	1.00	-79.91	0.08	-79.99	-79.98	-79.96	-79.85	-79.76
TempS	372	0.99	27.61	2.92	8.40	25.90	28.00	29.70	36.70
SalinityS	913	0.97	17.58	5.21	0.00	14.00	18.00	21.00	28.00
Year	0	1.00	1998.07	12.00	1980.00	1989.00	1996.00	2006.00	2023.00
Month	0	1.00	5.81	0.86	1.00	5.00	6.00	6.00	12.00

4.5.1.2 Oyster Landing Seines

Table 4.12: Brown shrimp (Oyster Landing) size summary

Table 4.12: Data summary

Name	sz_ol_brn
Number of rows	15212
Number of columns	12
Column type frequency:	
character	5
Date	1
numeric	6
Group variables	None

Variable type: character

Table 4.13: Brown shrimp (Oyster Landing) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Protocol	0	1	7	8	0	3	0
WQ.Source	0	1	8	19	0	4	0
WQ.Time	0	1	0	5	409	48	0
Species	0	1	5	5	0	1	0
Rep	0	1	4	6	0	100	0

Variable type: Date

Table 4.14: Brown shrimp (Oyster Landing) size summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1984-05-14	2023-08-29	1999-06-11	410

Variable type: numeric

Table 4.15: Brown shrimp (Oyster Landing) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Sample	0	1.00	384.06	230.34	10.0	185	383.0	543	891.0
Year	0	1.00	1999.55	10.18	1984.0	1991	1999.0	2006	2023.0
Month	0	1.00	6.27	1.50	1.0	5	6.0	7	12.0
Temp	895	0.94	25.64	2.82	6.6	24	26.0	28	31.4
Sal	903	0.94	32.54	3.53	7.7	31	33.8	35	38.7
Length	0	1.00	21.20	6.70	0.0	16	21.0	26	71.0

4.5.2 White Shrimp

4.5.2.1 SCDNR Creek Trawls

Table 4.16: White shrimp (SC) size summary

Table 4.16: Data summary

Name	sz_sc_wht
Number of rows	46792
Number of columns	14
Column type frequency:	
character	4
Date	1
numeric	9
Group variables	None

Variable type: character

Table 4.17: White shrimp (SC) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
DTStart	0	1	8	10	0	467	0
StationCode	0	1	4	4	0	7	0
SpCode	0	1	4	4	0	1	0
Species	0	1	5	5	0	1	0

Variable type: Date

Table 4.18: White shrimp (SC) size summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1980-05-08	2023-12-11	2003-07-29	467

Variable type: numeric

Table 4.19: White shrimp (SC) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
X	0	1.00	45299.88	23170.58	7.00	25467.75	48030.50	65043.25	81764.00
Coll	0	1.00	20031152.32	124812.02	19801003.00	19921067.00	20031027.00	20151040.00	20231080.00
Length	1	1.00	53.26	21.32	8.00	37.00	51.00	66.00	152.00
Lat	0	1.00	32.84	0.07	32.75	32.80	32.81	32.86	32.95
Long	0	1.00	-79.90	0.08	-79.99	-79.98	-79.96	-79.85	-79.76
TempS	215	1.00	29.22	2.98	11.00	28.30	29.60	31.00	36.70
SalinityS	822	0.98	15.99	5.81	0.00	12.00	17.00	20.00	30.00
Year	0	1.00	2003.02	12.48	1980.00	1992.00	2003.00	2015.00	2023.00
Month	0	1.00	7.43	1.11	1.00	7.00	7.00	8.00	12.00

4.5.2.2 Oyster Landing Seines

Table 4.20: White shrimp (Oyster Landing) size summary

Table 4.20: Data summary

Name	sz_ol_wht
Number of rows	24387
Number of columns	12

Column type frequency:	
character	5
Date	1
numeric	6
Group variables	None

Variable type: character

Table 4.21: White shrimp (Oyster Landing) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Protocol	0	1	7	8	0	3	0
WQ.Source	0	1	8	19	0	4	0
WQ.Time	0	1	0	5	874	51	0
Species	0	1	5	5	0	1	0
Rep	0	1	4	6	0	100	0

Variable type: Date

Table 4.22: White shrimp (Oyster Landing) size summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1984-07-12	2023-12-11	2000-09-11	436

Variable type: numeric

Table 4.23: White shrimp (Oyster Landing) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Sample	0	1.00	421.82	222.99	14.0	236.0	414	592.0	898.0
Year	0	1.00	2001.01	9.94	1984.0	1993.0	2000	2008.0	2023.0
Month	0	1.00	8.45	1.39	1.0	7.0	8	10.0	12.0
Temp	2200	0.91	25.85	3.78	10.6	23.8	27	28.8	31.4
Sal	2305	0.91	32.81	3.96	5.4	31.8	34	35.1	38.7
Length	0	1.00	23.84	7.09	3.0	19.0	23	29.0	71.0

4.6 About

Input `.qmd` file for this section was: **raw__juv.qmd** and can be found in the main directory.

Input data file(s) - as-used; can be found in `data/raw/juv`:

- `SCDNR_CreekTrawlShrimpCPUE.csv`
- `SCDNR_CreekTrawlShrimpSize.csv`
- `Oyster Landing Seine Shrimp Dataset - 1984-2023 (BWP, 2025-04-24)(kac 2025-06-02).csv`

Modifications to data:

- All files: changed `Species` from provided scientific names to 'brown', 'white', and 'pink'.

- Oyster Landing file: removed 'DataGap' from temp, salinity, and weight columns and turned these to numeric. Split data into counts data frame and sizes data frame. Pivoted sizes to long format.

Generated data file(s): Data frames were modified as described above, and written to `data/intermediate/juvenile_dfs.RData` (the single `.RData` file contains multiple data frames):

- `juv_sc_cpue`
- `juv_sc_size`
- `juv_ol_count`
- `juv_ol_size`

Other notes about data:

- Oyster Landing file used had a manual correction from Kim; original file had a typo - Sample 748, from 2017-07-20, LEN21, value of 221. Changed to 21.

5 Subadults

5.1 Datasets

There are two datasets that represent subadults: GADNR's EMTS sampling and SCDNR's Estuarine Trawls. We will explore each.

5.2 Graphics - Size Distributions

South Carolina's file has over 300,000 points, which makes for a beeswarm plot that is very large and slow to render. Georgia's file is "only" 65k and even it is too slow to render beeswarm plots. So rather than graphing the full datasets, I have randomly sampled 10,000 rows from GA and 15,000 rows from SC (because SC has three species).

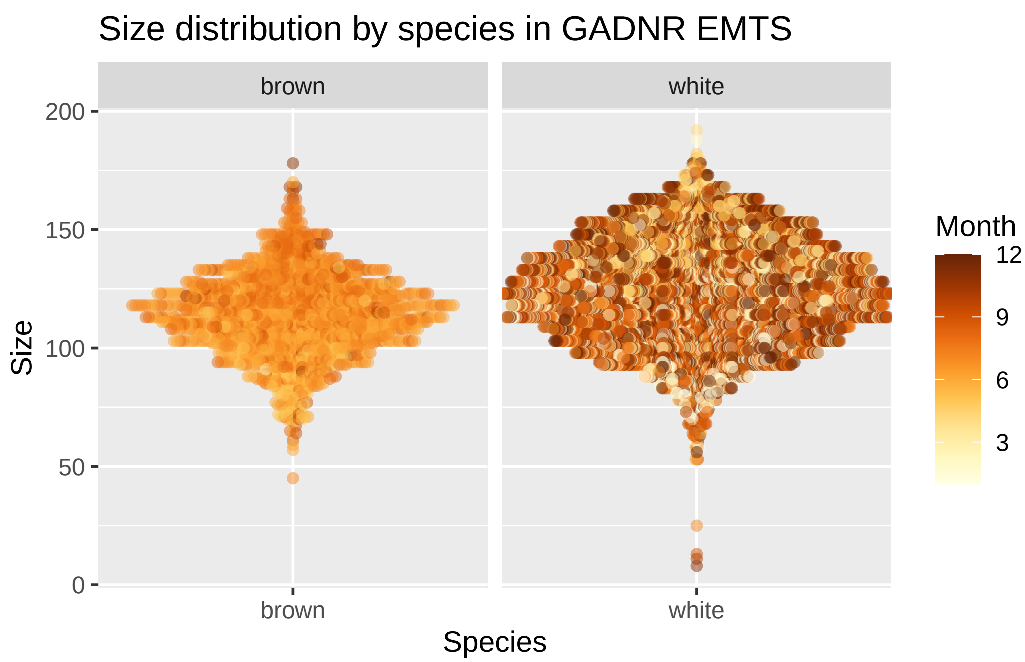


Figure 5.1: Beeswarm plot of subadult shrimp length for the GADNR data. Points represent measurements of individual shrimp, and are colored by the month in which they were captured. The y-axis represents length. More points spread out along the x-axis for a given length means that there were more individuals of that length captured than regions where points remain closer to the center.

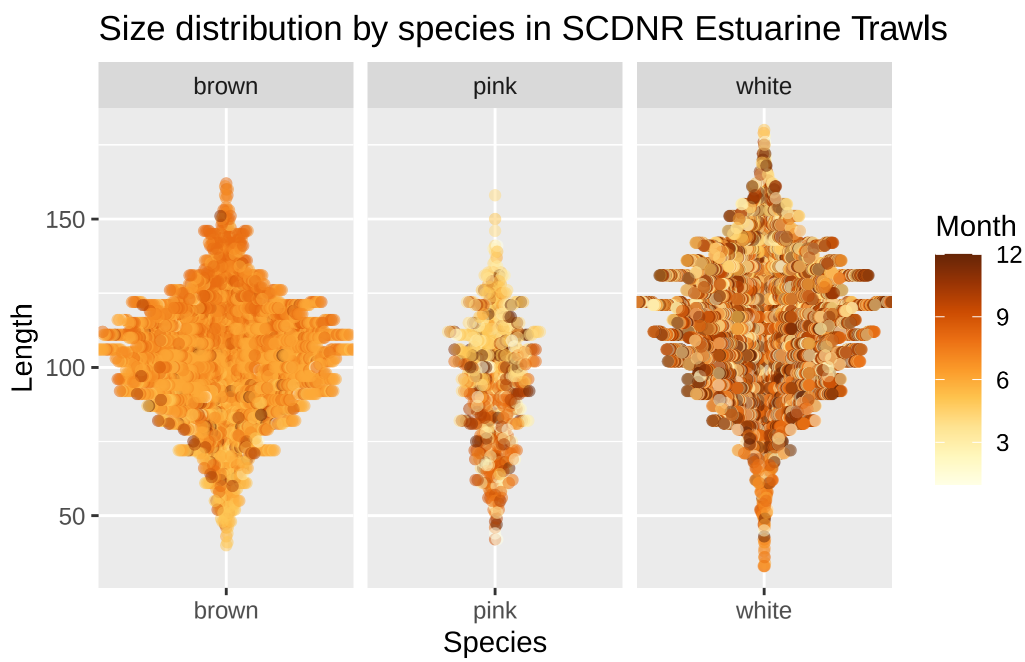


Figure 5.2: Beeswarm plot of subadult shrimp length for the SCDNR data. Points represent measurements of individual shrimp, and are colored by the month in which they were captured. The y-axis represents length. More points spread out along the x-axis for a given length means that there were more individuals of that length captured than regions where points remain closer to the center.

5.3 Graphics - Abundance/CPUE Boxplots

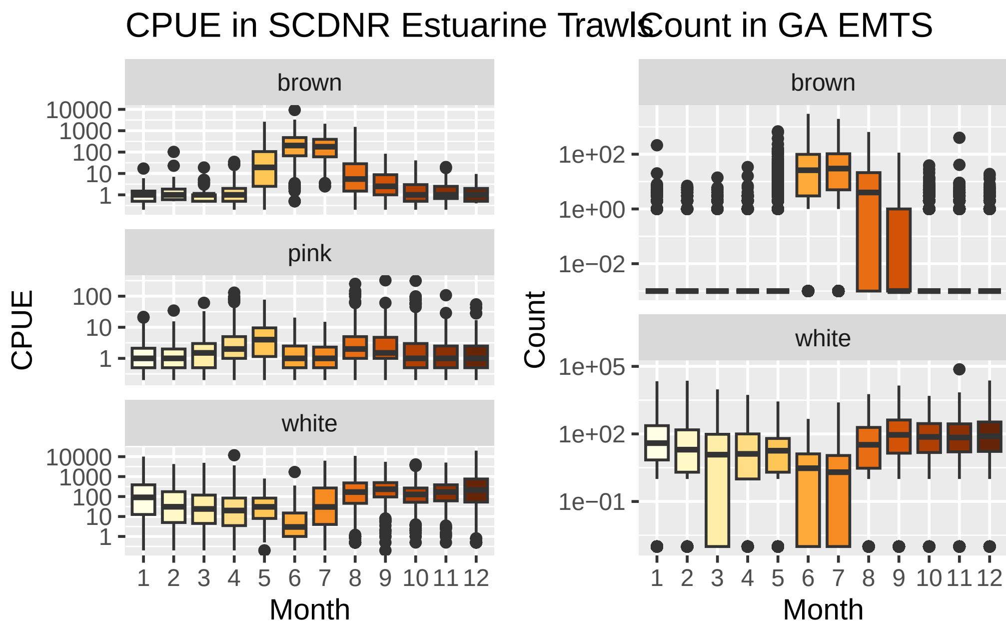


Figure 5.3: Boxplots representing shrimp abundance (as either CPUE or count, depending on the survey) by month. Note the log10-scaled y-axes.

Both surveys (unsurprisingly) show the same temporal pattern of abundance - brown shrimp most abundant in June and July, also high in August; and May in SC. For white shrimp, we see a dip in June and (less so) July.

5.4 Tabular summaries of abundance

5.4.0.1 GADNR EMTS

Table 5.1: Abundance (GA) data frame summary

Table 5.1: Data summary

Name	subad_ga_count
Number of rows	39669
Number of columns	14

Column type frequency:	
character	2
Date	1
numeric	11
Group variables	None

Variable type: character

Table 5.2: Abundance (GA) data frame summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
RefNum	0	1	12	12	0	19828	0
Species	0	1	5	5	0	2	0

Variable type: Date

Table 5.3: Abundance (GA) data frame summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
GuessedDate	0	1	1975-12-19	2021-10-26	2000-01-10	3254

Variable type: numeric

Table 5.4: Abundance (GA) data frame summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
TotWt	3	1.00	1491.00	8839.01	0	0.00	27.22	680.39	1394789.25
TotNum	83	1.00	115.10	610.81	0	0.00	2.00	43.00	73800.00
SampleWt	13	1.00	382.37	716.10	0	0.00	27.22	680.39	65770.89
SampleNum	120	1.00	28.03	47.17	0	0.00	2.00	42.00	1220.00
NumMeas	21154	0.47	12.23	16.40	0	0.00	2.00	30.00	207.00
LbsperHr	12269	0.69	17.83	75.69	0	0.16	2.08	12.88	7060.50
NumperLb	12359	0.69	33.89	253.35	0	15.27	25.60	40.00	31751.50
NumFemales	2395	0.94	7.63	10.59	0	0.00	2.00	15.00	122.00
NumMales	2382	0.94	6.27	9.46	0	0.00	1.00	11.00	97.00
Year	0	1.00	1998.94	13.00	1975	1988.00	2000.00	2010.00	2021.00
Month	0	1.00	6.49	3.42	1	4.00	6.00	9.00	12.00

5.4.0.2 SCDNR Estuarine Trawls

Table 5.5: Abundance (SC) data frame summary

Table 5.5: Data summary

Name	subad_sc_cpue
Number of rows	21771
Number of columns	13
Column type frequency:	
character	5
numeric	8
Group variables	None

Variable type: character

Table 5.6: Abundance (SC) data frame summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
estuary	0	1	2	2	0	6	0
StationCode	0	1	4	4	0	24	0
DTStart	0	1	8	10	0	1647	0
SpCode	0	1	4	4	0	3	0
Species	0	1	4	5	0	3	0

Variable type: numeric

Table 5.7: Abundance (SC) data frame summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
X	0	1	10886.00	6284.89	1.000e+00	5443.50	10886.00	16328.50	21771.00
Coll	0	1	19986283.95	122808.61	1.979e+07	19890034.00	19970150.00	20090016.00	20230114.00
Year	0	1	1998.61	12.29	1.979e+03	1989.00	1997.00	2009.00	2023.00
Month	0	1	6.60	3.37	1.000e+00	4.00	6.00	9.00	12.00
Day	0	1	17.04	8.02	1.000e+00	11.00	18.00	23.00	31.00
CPUE	4	1	119.11	525.17	0.000e+00	0.00	0.50	24.00	20052.00
Latitude	0	1	32.57	0.23	3.215e+01	32.32	32.67	32.77	32.83
Longitude	0	1	-80.30	0.36	-8.085e+01	-80.65	-80.29	-79.92	-79.89

5.5 Tabular summaries of size

5.5.1 Brown Shrimp

5.5.1.1 GADNR EMTS

Table 5.8: Brown shrimp (GA) size summary

Table 5.8: Data summary

Name	subad_ga_size_brown
Number of rows	11029
Number of columns	4
Column type frequency:	
character	3
numeric	1
Group variables	None

Variable type: character

Table 5.9: Brown shrimp (GA) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
RefNum	0	1	12	12	0	745	0
TowDate	0	1	8	10	0	308	0
Species	0	1	5	5	0	1	0

Variable type: numeric

Table 5.10: Brown shrimp (GA) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Size	0	1	113.84	17.45	8	103	114	125	182

5.5.1.2 SCDNR Estuarine Trawls

Table 5.11: Brown shrimp (SC) size summary

Table 5.11: Data summary

Name	sz_sc_brn
Number of rows	66499
Number of columns	15
Column type frequency:	
character	5
numeric	10
Group variables	None

Variable type: character

Table 5.12: Brown shrimp (SC) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
SpCode	0	1	4	4	0	1	0
Sex	0	1	0	1	61324	4	0
estuary	0	1	2	2	0	6	0
StationCode	0	1	4	4	0	24	0
Species	0	1	5	5	0	1	0

Variable type: numeric

Table 5.13: Brown shrimp (SC) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
X	0	1.00	129622.34	80691.21	1.000e+00	59387.50	126106.00	187401.50	304572.00
Coll	0	1.00	19955732.81	152429.92	1.979e+07	19870163.00	19940120.00	20010143.00	20230106.00
Length	0	1.00	102.12	19.99	2.100e+01	90.00	103.00	115.00	177.00
Year	0	1.00	1995.56	11.25	1.979e+03	1987.00	1994.00	2001.00	2023.00
Month	0	1.00	6.66	1.22	1.000e+00	6.00	6.00	7.00	12.00
Day	0	1.00	17.19	7.95	1.000e+00	10.00	19.00	24.00	31.00
Latitude	0	1.00	32.68	0.19	3.215e+01	32.67	32.77	32.80	32.83
Longitude	0	1.00	-80.12	0.30	-8.085e+01	-80.29	-79.97	-79.92	-79.89
TempB	1640	0.98	27.81	2.84	0.000e+00	27.00	28.40	29.40	32.40
SalinityB	2759	0.96	24.83	6.13	0.000e+00	21.00	26.00	30.00	39.00

5.5.2 White Shrimp

5.5.2.1 GADNR EMTS

Table 5.14: White shrimp (GA) size summary

Table 5.14: Data summary

Name	subad_ga_size_white
Number of rows	54505
Number of columns	4
Column type frequency:	
character	3
numeric	1
Group variables	None

Variable type: character

Table 5.15: White shrimp (GA) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
RefNum	0	1	12	12	0	2231	0
TowDate	0	1	8	10	0	726	0
Species	0	1	5	5	0	1	0

Variable type: numeric

Table 5.16: White shrimp (GA) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Size	0	1	124.02	21.61	2	108	124	140	212

5.5.2.2 SCDNR Estuarine Trawls

Table 5.17: White shrimp (SC) size summary

Table 5.17: Data summary

Name	sz_sc_wht
Number of rows	224834
Number of columns	15
Column type frequency:	
character	5
numeric	10
Group variables	None

Variable type: character

Table 5.18: White shrimp (SC) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
SpCode	0	1	4	4	0	1	0
Sex	0	1	0	1	173246	4	0
estuary	0	1	2	2	0	6	0
StationCode	0	1	4	4	0	24	0
Species	0	1	5	5	0	1	0

Variable type: numeric

Table 5.19: White shrimp (SC) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
X	0	1.00	165616.57	89820.94	7.000e+00	89692.25	172903.50	244450.75	307960.00
Coll	0	1.00	19991381.12	123497.97	1.979e+07	19890293.00	19970176.00	20090085.00	20230114.00
Length	0	1.00	113.38	22.51	1.100e+01	98.00	114.00	130.00	207.00
Year	0	1.00	1999.12	12.36	1.979e+03	1989.00	1997.00	2009.00	2023.00
Month	0	1.00	7.49	3.49	1.000e+00	4.00	8.00	10.00	12.00
Day	0	1.00	16.44	8.29	1.000e+00	10.00	17.00	23.00	31.00
Latitude	0	1.00	32.59	0.22	3.215e+01	32.46	32.67	32.77	32.83
Longitude	0	1.00	-80.27	0.35	-8.085e+01	-80.54	-80.24	-79.92	-79.89
TempB	4077	0.98	20.86	6.76	0.000e+00	15.00	20.60	27.90	32.40
SalinityB	4665	0.98	25.93	7.06	0.000e+00	22.00	27.00	31.00	209.00

5.6 About

Input .qmd file for this section was: **raw_subad.qmd** and can be found in the main directory.

Input data file(s) - as-used; can be found in data/raw/subad:

- GADNR_EMTS_PenaeidShrimp_Brown_Count_thru2021.xlsx
- GADNR_EMTS_PenaeidShrimp_White_Count_thru2021.xlsx
- GADNR_EMTS_PenaeidShrimp_SIZE_thru2021.xlsx, sheet = "Brown shrimp"
- GADNR_EMTS_PenaeidShrimp_SIZE_thru2021.xlsx, sheet = "White Shrimp"
- SCDNR_EstuarineTrawlShrimpCPUE.csv
- SCDNR_EstuarineTrawlShrimpSize.csv

Modifications to data:

- All files: changed Species from provided scientific names to 'brown', 'white', and 'pink'.

- GA counts file: extracted date from the Reference Number, as date was not its own field.

- SC size file: filtered data to remove lengths >1000. There were 3 such rows.

Generated data file(s): Data frames were modified as described above, and written to data/intermediate/subadult_dfs.RData (the single .RData file contains multiple data frames):

- subad_ga_count
- subad_ga_size
- subad_sc_cpue
- subad_sc_size

Other notes about data:

- In GA count files, early exploration not shown here indicated that NumperLb was miscalculated when the sample weight wasn't the typical 1360.78 - calculations in the spreadsheet used that value anyway. This issue seemed to happen when catch was low.

6 Adults - fisheries-independent

SEAMAP info here

About the Coastal Trawl Survey: https://seamap.org/seamap-sa-coastal-trawl/

Some excerpts:

- The standardized gear used since 1987 is a pair of 75 ft (22.9 m) mongoose-type Falcon trawl nets (manufactured by Beaufort Marine Supply, Beaufort, SC) without Turtle Excluder Devices. The body of the trawl net was constructed of #15 net twine with 47 mm stretch mesh.

- Historically, the catch from each net was processed separately and assigned a unique collection number, with data from both nets at each station pooled for analysis to form a standard unit of effort (tow). Beginning in 2021, only the catch from one net was processed with the tail bag of the other net fished untied.

- three sampling seasons: Spring (April - May), Summer (July - August), and Fall (September - November).

So it will probably be best to average to the event, because we've got events in 2022 with only one collection (prior to 2022, it was 2 collections per event).

Starting in 1989, which is when this file starts, sampling was pretty consistent - seasonally rather than monthly; daytime; 75ft nets.

There's also some inner vs. outer depth zone stuff to check. See below - 14,576 rows are from "inner" and only 534 from "outer", so will remove "outer" completely.

More on depth zones, from the Figure 1 caption on the Coastal Trawl website linked above: "The first map shows what's going on, and has this in the caption:"….Inner (shallow) strata are sampled during all seasons (1990-present). Outer (deep) strata were sampled south in Spring (green) and north in Fall (yellow) from 1990-2000. In 1989, both depth zones were sampled over the entire latitudinal range in Spring and Fall."

NC, SC, GA, and FL are present in this dataset.

6.1 Datasets

There are two SEAMAP files: one of events, and one of catch data.

Looks like when no shrimp were caught in a net, the species is left blank and count and weight are both 0. So I'll widen the data frame, filling in 0s for NAs. Then lengthen back out so 0s are all accounted for.

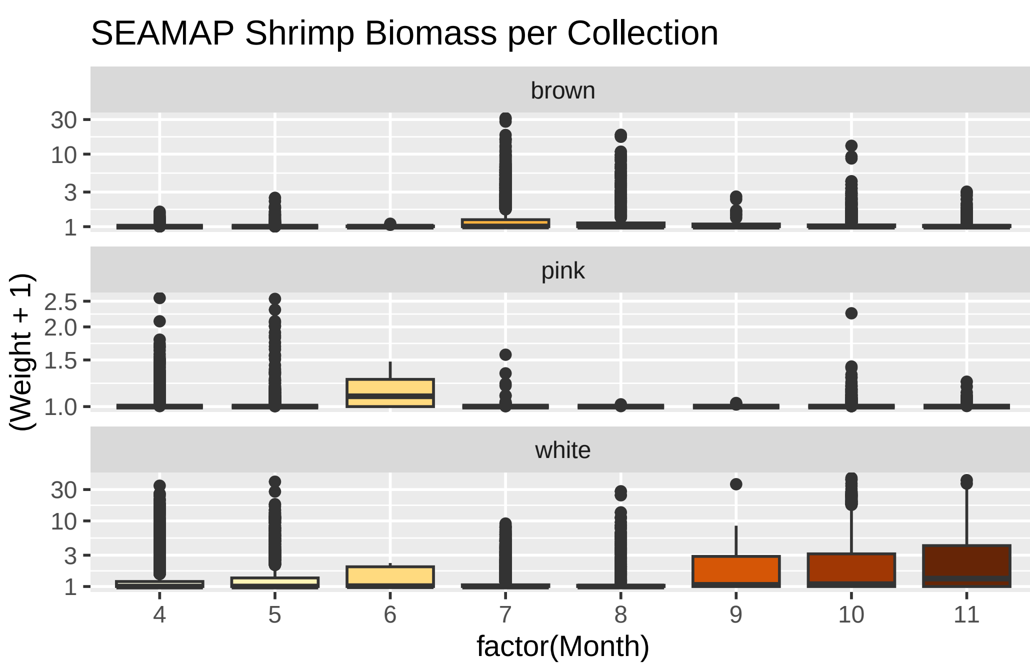
Also filtering to "INNER" depth zone.

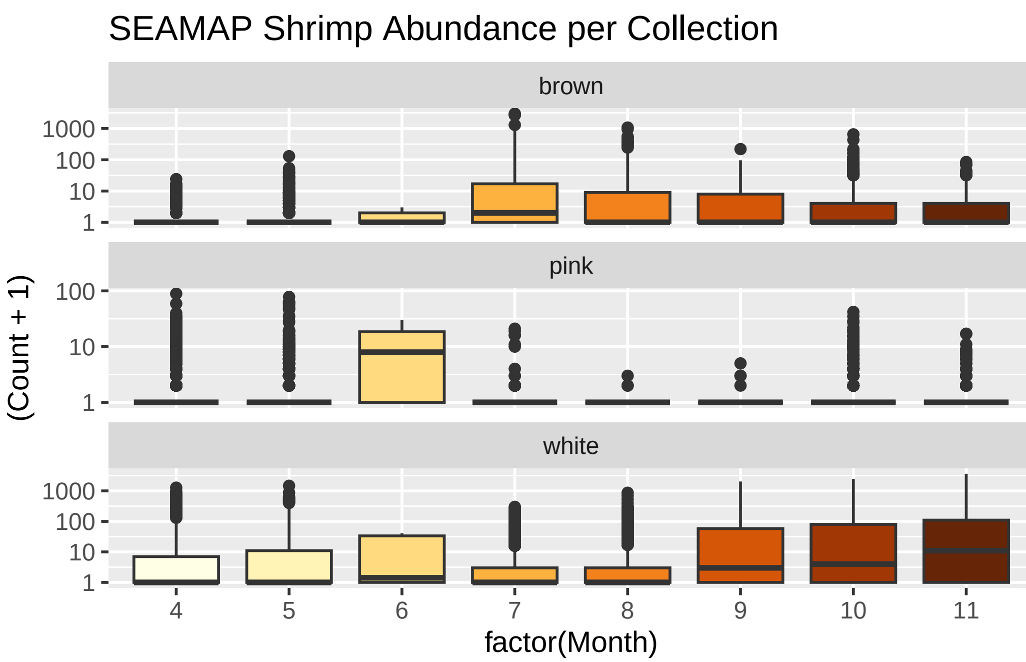
Weight and count were both recorded, even in subsamples, BUT "when large numbers of a species occurred in the processed catch", all individuals were weighed; a subsample was processed for abundance (typically 32-64 individuals); and count was estimated from the weight.

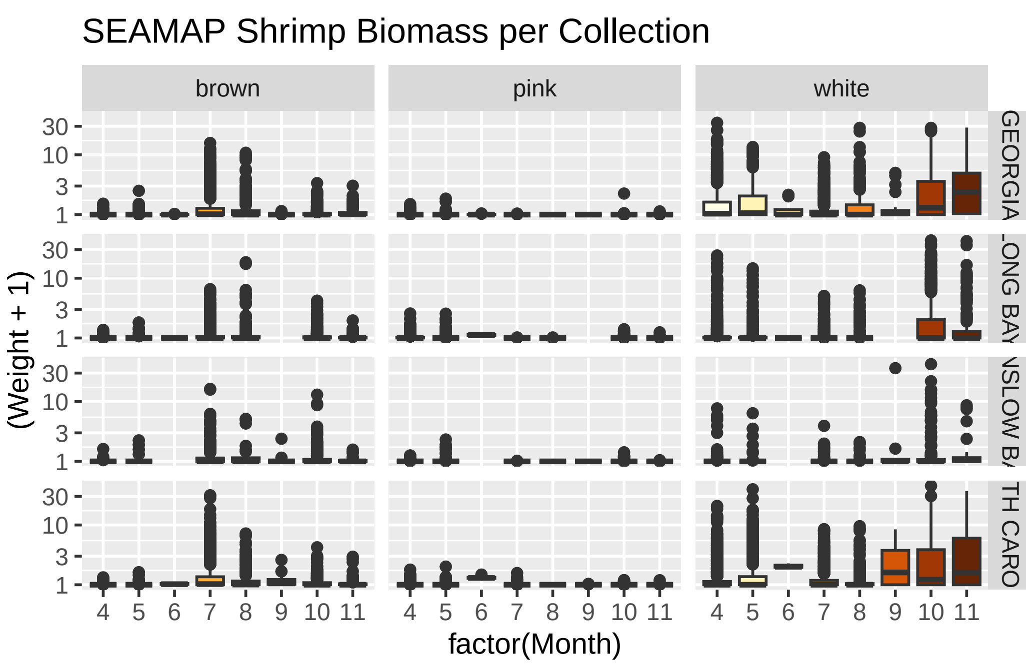
FOR THAT REASON I will use weights in my analyses, because it is the most likely piece to have been measured rather than estimated.

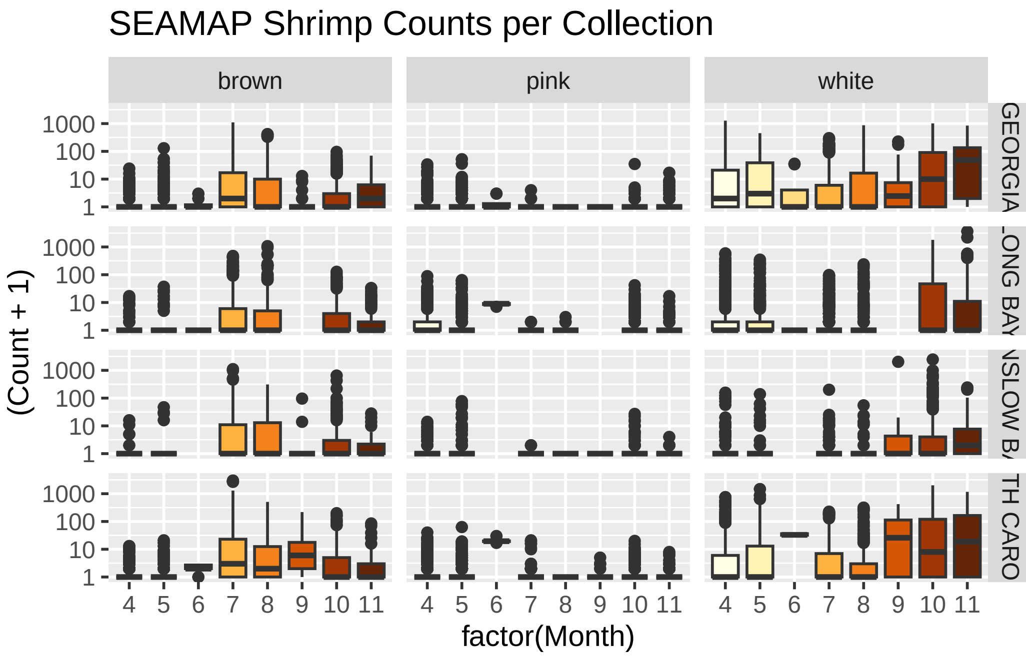
6.2 Graphics - Abundance/Landings/CPUE Boxplots

STILL IN PROGRESS




SEAMAP Shrimp Abundance per Collection
brown
pink
white
(Count + 1)
1000
100
10
1
4
5
6
7
8
9
10
11
factor(Month)


SEAMAP Shrimp Biomass per Collection
brown
pink
white
GEORGIA
(Weight + 1)
30
10
3
1
4
5
6
7
8
9
10
11
factor(Month)



6.3 Tabular summaries of abundance

6.3.0.1 Brown Shrimp

Table 6.1: Brown shrimp SEAMAP summary

Table 6.1: Data summary

Name	filter(adsm_long, species...
Number of rows	11310
Number of columns	9
Column type frequency:	
character	2
Date	1
numeric	6
Group variables	None

Variable type: character

Table 6.2: Brown shrimp SEAMAP summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Region	0	1	7	14	0	4	0
species	0	1	5	5	0	1	0

Variable type: Date

Table 6.3: Brown shrimp SEAMAP summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1989-04-12	2022-11-16	2005-07-19	994

Variable type: numeric

Table 6.4: Brown shrimp SEAMAP summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Year	0	1	2004.80	9.08	1989	1997	2005	2012.00	2022.0
Month	0	1	7.30	2.41	4	5	7	10.00	11.0
Event	0	1	2005105.07	9090.43	1989001	1997302	2005309	2012542.50	2022559.0
Collection	0	1	20048332.60	90798.12	19890001	19970302	20050310	20120542.75	20220559.0
Weight	0	1	0.18	0.92	0	0	0	0.03	30.4
Count	0	1	11.64	69.76	0	0	0	2.00	2983.0

6.3.0.2 White Shrimp

Table 6.5: White shrimp SEAMAP summary

Table 6.5: Data summary

Name	filter(adsm_long, species...
Number of rows	11310
Number of columns	9
Column type frequency:	
character	2
Date	1
numeric	6

Group variables	None

Variable type: character

Table 6.6: White shrimp SEAMAP summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Region	0	1	7	14	0	4	0
species	0	1	5	5	0	1	0

Variable type: Date

Table 6.7: White shrimp SEAMAP summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1989-04-12	2022-11-16	2005-07-19	994

Variable type: numeric

Table 6.8: White shrimp SEAMAP summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Year	0	1	2004.80	9.08	1989	1997	2005	2012.00	2022.00
Month	0	1	7.30	2.41	4	5	7	10.00	11.00
Event	0	1	2005105.07	9090.43	1989001	1997302	2005309	2012542.50	2022559.00
Collection	0	1	20048332.60	90798.12	19890001	19970302	20050310	20120542.75	20220559.00
Weight	1	1	1.00	2.86	0	0	0	0.46	44.07
Count	0	1	36.95	116.85	0	0	0	15.00	3641.00

6.4 About

Input `.qmd` file for this section was: **raw_ad_seamap.qmd** and can be found in the main directory.

Input data file(s) - as-used; can be found in `data/raw/ad_seamap`:

- `SEAMAP_ShrimpCatches_19892022.csv`

Modifications to data:

- Changed `Species` from provided names to 'brown', 'white', and 'pink'.

- For `adsm_long` and `adsm_wide` data frames, filtered to only the "INNER" depth zone and filled in NAs with 0s, because when nothing was caught in a net, both columns in the original data were left blank.

Generated data file(s): Data frames were modified as described above, and written to `data/intermediate/adultSeamap_dfs.RData` (the single `.RData` file contains multiple data frames):

- `seamap` - the unfiltered data frame
- `adsm_wide`
- `adsm_long`

Other notes about data:

- Contains 4 states: GA, NC, SC, FL

7 Adults - landings and trip tickets

Commercial data here

7.1 Datasets

There are three files that represent commercial fisheries data for adult shrimp: Landings, from NMFS, at two different frequencies (annual for both species, and for white shrimp, fall vs. winter-spring-summer); and trip tickets, by 2-month blocks for both species.

For brown shrimp, calendar year is fine. White shrimp have different life cycles, and we are interested in spawning adults vs. non-spawning adults for each ‘shrimp year’; so we obtained the head-off white shrimp data file, split into seasons.

7.2 Graphics - Abundance/Landings/CPUE Boxplots

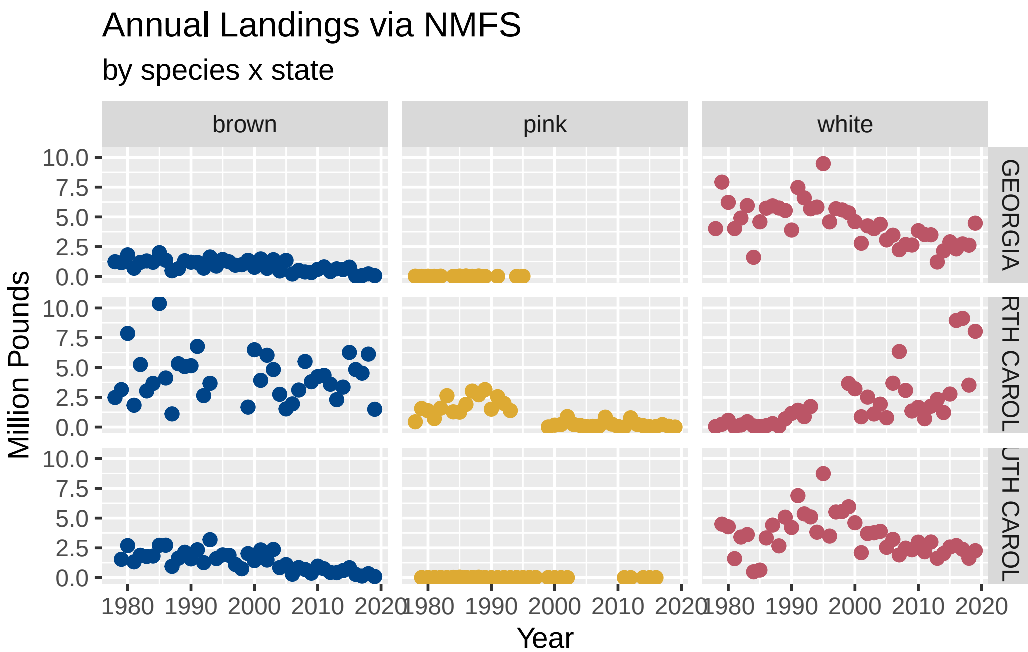


Figure 7.1: Time series plots of annual landings by species and state.

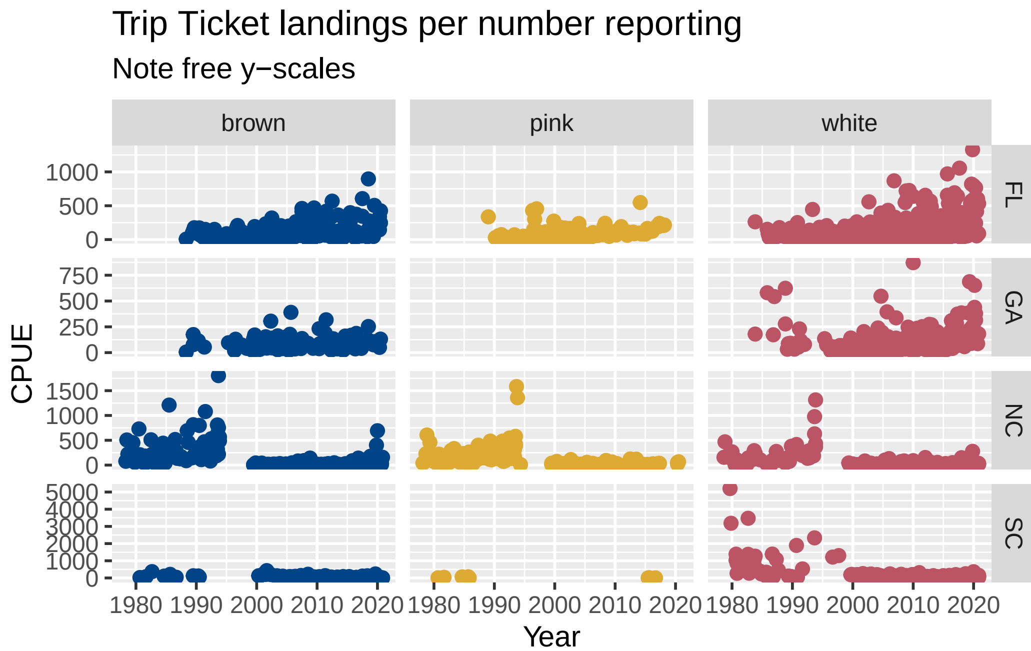


Figure 7.2: Time series plots of estimated CPUE by species and state, from trip ticket data. This plot uses free y-axes for overall context. Large differences probably signal changes in trip ticket collection and reporting, as states are in charge of their programs and may implement changes.

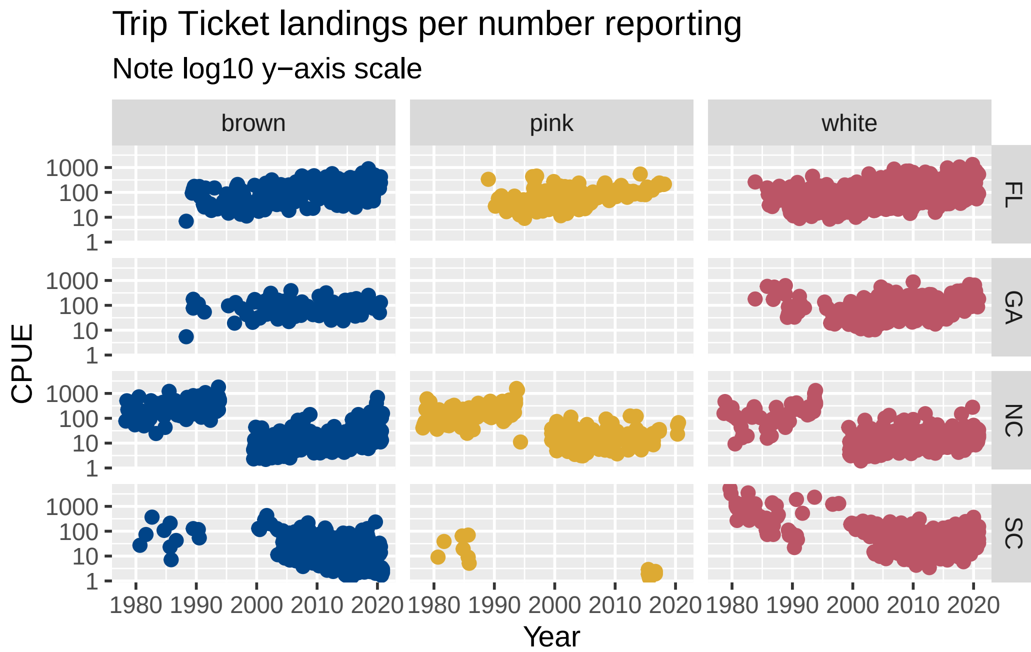


Figure 7.3: Time series plots of estimated CPUE by species and state, from trip ticket data. This plot is the same as that above, but it uses log-10 axes for easier comparison, especially at the low abundances. Large differences probably signal changes in trip ticket collection and reporting, as states are in charge of their programs and may implement changes.

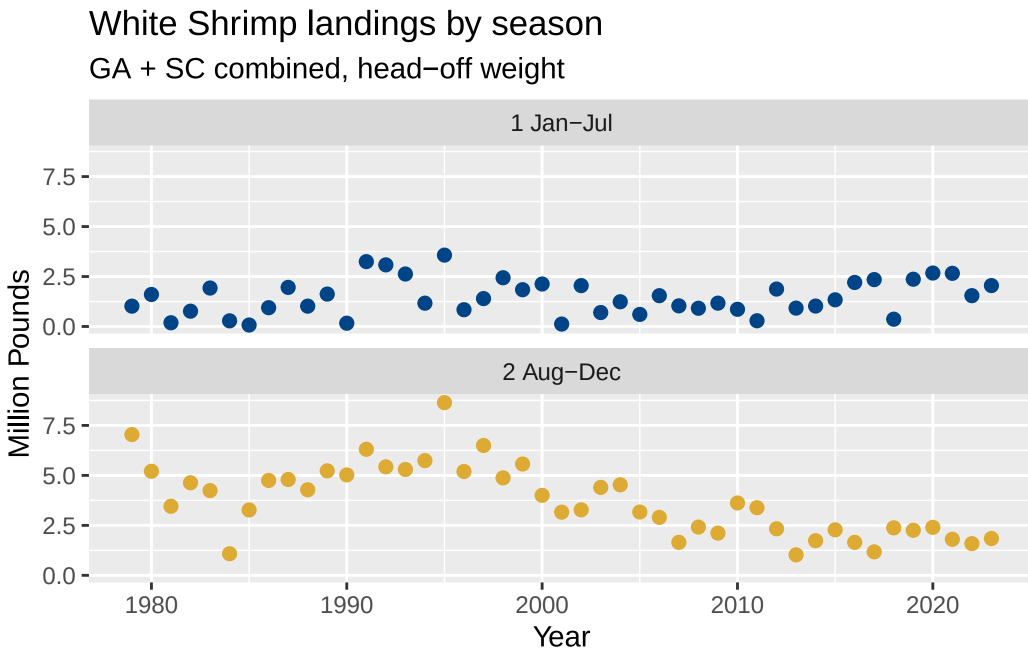


Figure 7.4: Time series plots of white shrimp landings, faceted by season (1 Jan-Jul represents spring spawning adults; 2 Aug-Dec represents fall non-spawning adults).

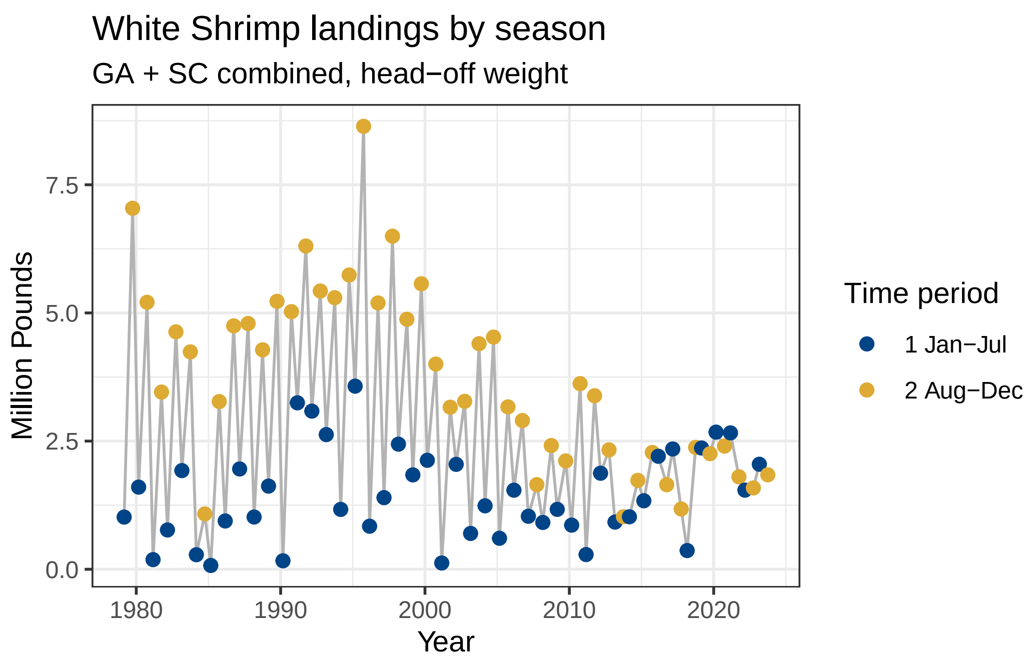


Figure 7.5: Time series plots of white shrimp landings, colored by season (1 Jan-Jul represents spring spawning adults; 2 Aug-Dec represents fall non-spawning adults) but in the same panel and connected with a line. Jan-Jul points have been placed at March 1st each year, and Aug-Dec points placed at October 1st of each year.

7.3 Tabular summaries of abundance

7.3.0.1 Annual Landings (NMFS)

Brown Shrimp

Table 7.1: Brown shrimp annual landings summary

Table 7.1: Data summary

Name	filter(landings_annual, S...
Number of rows	120
Number of columns	10
Column type frequency:	
character	5
numeric	5

Group variables	None

Variable type: character

Table 7.2: Brown shrimp annual landings summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
State	0	1	7	14	0	3	0
NMFS.Name	0	1	23	23	0	1	0
Collection	0	1	10	10	0	1	0
Confidentiality	0	1	6	6	0	1	0
Species	0	1	5	5	0	1	0

Variable type: numeric

Table 7.3: Brown shrimp annual landings summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Year	0	1	1998.78	12.31	1978.00	1988.00	1999.50	2009.25	2019.00
Pounds	0	1	2059941.38	1872035.54	41907.00	765917.75	1427067.50	2720053.75	10378157.00
Dollars	0	1	4039632.03	3692856.60	59049.00	1461823.75	2876552.00	4983011.25	18847514.00
Millions_of_dollars	0	1	4.04	3.69	0.06	1.46	2.88	4.98	18.85
Millions_of_Pounds	0	1	2.06	1.87	0.04	0.77	1.43	2.72	10.38

White Shrimp

Table 7.4: White shrimp annual landings summary

Table 7.4: Data summary

Name	filter(landings_annual, S…
Number of rows	120
Number of columns	10
Column type frequency:	
character	5
numeric	5

Group variables	None

Variable type: character

Table 7.5: White shrimp annual landings summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
State	0	1	7	14	0	3	0
NMFS.Name	0	1	23	23	0	1	0
Collection	0	1	10	10	0	1	0
Confidentiality	0	1	6	6	0	1	0
Species	0	1	5	5	0	1	0

Variable type: numeric

Table 7.6: White shrimp annual landings summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Year	0	1	1998.78	12.31	1978.00	1988.00	1999.50	2009.25	2019.00
Pounds	0	1	3372708.22	2171286.61	11372.00	1743573.75	3144572.50	4590372.75	9472533.00
Dollars	0	1	8334883.54	5549245.49	25119.00	4402430.75	7460425.00	11911377.25	23688259.00
Millions_of_dollars	0	1	8.33	5.55	0.03	4.40	7.46	11.91	23.69
Millions_of_Pounds	0	1	3.37	2.17	0.01	1.74	3.14	4.59	9.47

7.3.0.2 Trip Ticket Landings

Brown Shrimp

Table 7.7: Brown shrimp trip tickets summary

Table 7.7: Data summary

Name	filter(tripTick_abund, Sp...
Number of rows	983
Number of columns	15
Column type frequency:	
character	5
numeric	10

Group variables	None

Variable type: character

Table 7.8: Brown shrimp trip tickets summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Area.name	0	1	7	46	0	13	0
Common.Name	0	1	14	14	0	1	0
Species	0	1	5	5	0	1	0
Wave	0	1	9	9	0	6	0
State	0	1	0	2	237	5	0

Variable type: numeric

Table 7.9: Brown shrimp trip tickets summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
object	0	1	1883.32	1036.12	2.00	1032.00	1952.00	2785.00	3573.00
Year	0	1	2004.77	10.94	1978.00	1998.00	2007.00	2014.00	2020.00
WAVE	0	1	3.91	1.24	1.00	3.00	4.00	5.00	6.00
Lat	0	1	32.18	1.93	28.53	30.50	32.24	34.27	35.43
Long	0	1	-78.66	1.75	-81.21	-80.35	-78.25	-77.36	-75.68
annland	0	1	287816.95	768368.26	24.00	10378.01	41119.21	167878.98	8084396.00
numdeal	0	1	14777.14	67470.92	3.00	161.00	900.00	4312.00	919803.00
numfisher	0	1	14777.14	67470.92	3.00	161.00	900.00	4312.00	919803.00
CPUE_approx	0	1	140.18	294.14	0.52	13.27	42.87	149.40	4847.00
Wave_start	0	1	6.81	2.49	1.00	5.00	7.00	9.00	11.00

White Shrimp

Table 7.10: White shrimp trip tickets summary

Table 7.10: Data summary

Name	filter(tripTick_abund, Sp...
Number of rows	1636

Number of columns	15
Column type frequency:	
character	5
numeric	10
Group variables	None

Variable type: character

Table 7.11: White shrimp trip tickets summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Area.name	0	1	7	46	0	14	0
Common.Name	0	1	14	14	0	1	0
Species	0	1	5	5	0	1	0
Wave	0	1	9	9	0	6	0
State	0	1	0	2	256	5	0

Variable type: numeric

Table 7.12: White shrimp trip tickets summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
object	0	1	1889.27	1018.10	6.00	1056.25	1938.50	2755.75	3575.00
Year	0	1	2004.90	10.65	1978.00	1998.00	2006.00	2014.00	2020.00
WAVE	0	1	3.87	1.68	1.00	3.00	4.00	5.00	6.00
Lat	0	1	31.77	1.92	25.25	30.50	31.33	33.19	35.43
Long	0	1	-79.26	1.76	-81.21	-80.50	-80.34	-78.25	-75.68
annland	0	1	349093.95	796531.34	32.24	29626.52	108025.30	314248.95	12391609.78
numdeal	0	1	15000.49	47907.43	16.00	309.75	1330.00	8266.50	648442.00
numfisher	0	1	15000.49	47907.43	16.00	309.75	1330.00	8266.50	648442.00
CPUE_approx	0	1	145.03	281.83	0.92	19.60	52.89	156.18	5202.00
Wave_start	0	1	6.75	3.35	1.00	5.00	7.00	9.00	11.00

7.3.0.3 Seasonal White Shrimp Landings

Table 7.13: White Shrimp (SC + GA) seasonal landings.

Table 7.13: Data summary

Name	landings_seas
Number of rows	90
Number of columns	6
Column type frequency:	
character	3
numeric	3
Group variables	None

Variable type: character

Table 7.14: White Shrimp (SC + GA) seasonal landings.

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
season	0	1	4	11	0	2	0
Species	0	1	5	5	0	1	0
Months	0	1	9	9	0	2	0

Variable type: numeric

Table 7.15: White Shrimp (SC + GA) seasonal landings.

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Year	0	1	2001.0	13.06	1979	1990	2001.0	2012	2023
lbs_headOff	0	1	2593141.3	1799455.64	74022	1169931	2164551.5	3542747	8639831
Period	0	1	1.5	0.50	1	1	1.5	2	2

7.4 About

Input `.qmd` file for this section was: **raw_ad_commercial.qmd** and can be found in the main directory.

Input data file(s) - as-used; can be found in `data/raw/ad_commercial`:

- `Nonconfidential Shrimp landings by species area year two month block.xlsx` - Trip Tickets. Commercial shrimp landings in NC, SC, GA, and FL, by 2-month block, as reported by commercial fishers and seafood processors. From state trip ticket data. per the data provider, this dataset probably best if we're interested in estimating CPUE. " Trip tickets, which have area fished, came online at different times (GA 1999, SC 2003, NC 1994). That is causing the differences. Also NC didn't always differentiate among the species (I think 96 to 98). "
- `Shrimp Landings NC SC GA from NMFS.csv` - both species, calendar year level (okay for brown shrimp, not white). Annual commercial shrimp landings in NC, SC, and GA, as reported by commercial fishers and seafood processors, and downloaded from https://www.fisheries.noaa.gov/foss/f?p=215:200:::::: per the data provider, this dataset is probably the better one for total landings.
- `scga_landings_headOff_whiteShrimp_byseason.csv` - white shrimp, by Fall vs. Winter-Spring-Summer. Combined white shrimp landings for SC and GA by two 'seasons'. The Fall is defined as Aug-Dec and the Win_Spr_Sum is defined as Jan-July. As such, these two seasons can be combined to come up with 'Annual' landings.

Modifications to data:

- All files: changed `Species` from provided names to 'brown', 'white', and 'pink'.

- Trip Tickets file: Calculated `CPUE_approx` as landings / number of dealers (number of dealers and number of fishers were the same). Made two new columns for two-month `WAVE`: `Wave` to identify the months, e.g. `1 Jan-Feb`, `2 Mar-Apr`, etc.; and `Wave_start` to identify the first month numerically, for better plotting of time series, e.g. Wave 1 (Jan-Feb) is assigned 1; Wave 2 (Mar-Apr) is assigned 3; etc. Added a column for `State`, based on mapping the lat/longs of each named location in the file during earlier exploration.

- Annual Landings from NMFS file: both `Pounds` and `Dollars` were read in as character due to commas in the number formatting; parsed these to numeric. Then calculated versions where the numbers will be smaller: `Millions_of_Dollars` and `Millions_of_Pounds`.

- Seasonal head-off landings of white shrimp: added two columns for better ordering, because 'Win_Spr_Sum' is Jan-July, and 'Fall' is Aug-Dec of each year. Added `Months` to show this, and `Period` where Fall is 1 and Win_Spr_Sum is 2. Otherwise Fall appears first in the data frame due to alphabetical ordering; but chronologically it should be 2nd.

Generated data file(s): Data frames were modified as described above, and written to `data/intermediate/adultCommercial_dfs.RData` (the single `.RData` file contains multiple data frames):

- `landings_annual`

- `landings_seas`
- `tripTick_abund`

Other notes about data:

- `landings_seas` only represents two states: GA and SC. The annual landings and trip ticket data frames also include NC.

8 Food sources (benthic cores)

Stuff here

Part III

Environmental Data Exploration

9 Water Temperature at Charleston Harbor

stuff

10 Salinity

from SCDNR trawls

Part IV

Summarizing to Abundance Index

11 Postlarval Stage

Datasets - .....

12 Juvenile

Datasets .....

Multiple datasets here; abundances and sizes also present

13 Subadults

Stuff here ....

14 Adults - fishery-independent

SEAMAP info here

14.1 test

rendering with freeze: auto

14.2 another test

```
[1] 4
```

15 Adults - landings and trip tickets

Commercial data here

15.1 test

code:

```
[1] 2
```

The end

16 Food sources (benthic cores)

Stuff here

Part V

Summarizing Environmental Data

17 Water Temperature at Charleston Harbor

stuff

18 Salinity

from SCDNR trawls

Part VI

Relationships between datasets

19 Temperature thresholds

some stuff

20 Nursery Period Salinity

stuff

21 Regression Models

Start to explore what happens when we do more than just correlations.